

MATH 3260 - Chapter 9 Notes: Linear Programming

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

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These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 9.1:** Linear inequalities, systems of inequalities, half-planes, feasible regions, and modeling
2. **Section 9.2:** Two-variable linear programming, the vertex principle, graphical optimization, multiple optima, and unbounded regions
3. **Section 9.3:** Standard form, slack variables, basic feasible solutions, simplex tableaus, pivoting, maximization, and exceptional cases
4. **Section 9.4:** Minimization, primal-dual construction, weak and strong duality, complementary slackness, and economic interpretation
5. **Section 9.5:** Mixed constraints, surplus and artificial variables, Phase I/Phase II, infeasibility, and transportation-style applications

Each section concludes with a terminology table, a worksheet, and complete instructor solutions.

Original-example policy

All numerical examples are original. Themes are used selectively from the San Antonio Spurs, Backstreet Boys, hiking, badminton, driving, travel, and R&B music. Every business, event, route, or production interpretation is a fictional teaching model.

Section 9.1 - Systems of Linear Inequalities

Learning objectives

Students should be able to:

- determine whether a point satisfies a linear inequality;
- graph a linear inequality using its boundary line and a test point;

- distinguish solid and dashed boundaries;
- graph a system of inequalities as an intersection of half-planes;
- compute and label the vertices of a feasible region;
- translate verbal resource restrictions into linear inequalities.

Graphical problems in this chapter are carried out in

$$\mathbb{R}^2$$

unless stated otherwise.

Definition: linear inequality in two variables

A linear inequality has one of the forms

$$ax + by < c, \quad ax + by \leq c, \quad ax + by > c, \quad ax + by \geq c,$$

where a and b are not both zero. A point (x, y) is a solution when substitution makes the inequality true.

Boundary line and the two half-planes

The equation

$$ax + by = c$$

is the **boundary line**. It divides the plane into two open half-planes:

$$ax + by < c \quad \text{and} \quad ax + by > c.$$

For a non-strict inequality, the boundary is included and is drawn as a solid line. For a strict inequality, the boundary is excluded and is drawn as a dashed line.

Why one test point determines the entire side

Write

$$L(x, y) = ax + by - c.$$

The boundary line is $L(x, y) = 0$, or

$$y = \frac{c - ax}{b}$$

when $b \neq 0$. The line divides the plane into two half-planes. Since L is continuous, its sign cannot change without crossing the boundary $L = 0$. Therefore one test point identifies which side satisfies the inequality.

Detailed example: graphing one inequality

Graph

$$2x - y \leq 4.$$

First replace $\leq$ by $=$:

$$2x - y = 4 \iff y = 2x - 4.$$

Because the inequality includes equality, draw the boundary as a solid line. Test the origin:

$$2(0) - 0 = 0 \leq 4.$$

Thus the solution is the side containing the origin. Solving directly for y confirms the direction:

$$2x - y \leq 4 \iff -y \leq 4 - 2x \iff y \geq 2x - 4.$$

The inequality reverses when multiplying by -1 .

Systems as intersections

A system requires every inequality to hold simultaneously. If the individual solution sets are $H_1, H_2, \dots, H_k$, then the feasible set is

$$F = H_1 \cap H_2 \cap \dots \cap H_k.$$

Each half-plane is convex, and an intersection of convex sets is convex. Therefore, every feasible region formed by linear inequalities is convex.

Proof that a feasible region is convex

Take any two feasible points $\mathbf{u}$ and $\mathbf{v}$, and let

$$\mathbf{w} = t\mathbf{u} + (1 - t)\mathbf{v}, \quad 0 \leq t \leq 1.$$

Suppose one constraint is

$$\mathbf{a}^T \mathbf{x} \leq b.$$

Because $\mathbf{u}$ and $\mathbf{v}$ are feasible,

$$\mathbf{a}^T \mathbf{u} \leq b, \quad \mathbf{a}^T \mathbf{v} \leq b.$$

Then

$$\begin{aligned} \mathbf{a}^T \mathbf{w} &= \mathbf{a}^T (t\mathbf{u} + (1 - t)\mathbf{v}) \\ &= t\mathbf{a}^T \mathbf{u} + (1 - t)\mathbf{a}^T \mathbf{v} \\ &\leq tb + (1 - t)b = b. \end{aligned}$$

Thus $\mathbf{w}$ satisfies every constraint and is feasible. Hence the entire segment joining two feasible points lies in the feasible region.

Detailed feasible-region example

Consider

$$\begin{aligned}x &\geq 0, \\y &\geq 0, \\x + y &\leq 8, \\2x + y &\leq 12, \\x + 2y &\leq 14.\end{aligned}$$

The vertices are found by intersecting boundary lines and retaining only feasible intersections.

- On the x -axis, $2x \leq 12$, so the farthest feasible point is $(6, 0)$.
- Intersect $x + y = 8$ and $2x + y = 12$:

$$\begin{aligned}x + y &= 8, \\2x + y &= 12.\end{aligned}$$

Subtracting gives $x = 4$, then $y = 4$.

- Intersect $x + y = 8$ and $x + 2y = 14$. Subtracting gives $y = 6$, then $x = 2$.
- On the y -axis, $2y \leq 14$, so the farthest feasible point is $(0, 7)$.

Therefore the vertices are

$$(0, 0), (6, 0), (4, 4), (2, 6), (0, 7).$$

Modeling example: hiking and badminton schedule

A fictional weekly activity plan uses

- x : number of long hiking sessions;
- y : number of badminton sessions.

Suppose:

- at most eight total sessions are available: $x + y \leq 8$;
- hiking uses two recovery units and badminton uses one, with at most twelve units: $2x + y \leq 12$;
- hiking uses one travel block and badminton uses two facility blocks, with at most fourteen blocks: $x + 2y \leq 14$;
- negative sessions are impossible: $x \geq 0, y \geq 0$.

This verbal model produces exactly the feasible region above. Linear programming begins by translating each sentence into one linear inequality.

Section 9.1 terminology

Term	Meaning
Linear inequality	A comparison involving a linear expression
Boundary line	The equation obtained by replacing the inequality symbol with equality
Half-plane	One of the two regions determined by a line
Feasible point	A point satisfying every constraint
Feasible region	The intersection of all constraint regions
Active constraint	A constraint satisfied with equality at a point

Term	Meaning
Vertex	A corner point formed by active boundaries
Bounded region	A feasible region contained inside a sufficiently large circle
Unbounded region	A feasible region extending indefinitely

Section 9.2 - Linear Programming in Two Variables

Learning objectives

Students should be able to:

- identify the decision variables, objective function, and constraints;
- locate the feasible region and all of its vertices;
- use the vertex method to maximize or minimize a linear objective;
- explain why a vertex contains an optimum when an optimum exists;
- recognize multiple optimal solutions, infeasibility, and unboundedness;
- formulate and solve a two-variable application.

Definition: linear programming problem

A linear programming problem optimizes a linear objective

$$z = c_1x + c_2y$$

subject to finitely many linear constraints. The decision variables describe controllable quantities, and the feasible region contains all choices satisfying every constraint.

Level lines of the objective

The equation

$$c_1x + c_2y = k$$

is a level line on which the objective has the constant value k . All level lines are parallel. Increasing or decreasing k slides the line without rotating it.

Graphically, optimization moves the level line in the improving direction until it last touches the feasible region.

The vertex principle

Theorem: optimal solution at a vertex

For a nonempty polygonal feasible region, if a linear objective has a finite maximum or minimum, then at least one optimal solution is a vertex. If two distinct feasible points are optimal, then every point on the segment joining them is also optimal.

Detailed proof of the vertex principle

Let the objective be

$$f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}.$$

Suppose an optimal point $\mathbf{x}^*$ is not a vertex. Then $\mathbf{x}^*$ lies in a nontrivial line segment contained in the feasible region, so there are distinct feasible points $\mathbf{u}$ and $\mathbf{v}$ and a number $0 < t < 1$ such that

$$\mathbf{x}^* = t\mathbf{u} + (1 - t)\mathbf{v}.$$

By linearity,

$$\begin{aligned}
f(\mathbf{x}^*) &= \mathbf{c}^T(t\mathbf{u} + (1-t)\mathbf{v}) \\
&= t\mathbf{c}^T\mathbf{u} + (1-t)\mathbf{c}^T\mathbf{v} \\
&= tf(\mathbf{u}) + (1-t)f(\mathbf{v}).
\end{aligned}$$

For a maximum, neither $f(\mathbf{u})$ nor $f(\mathbf{v})$ can exceed $f(\mathbf{x}^*)$. A weighted average of two numbers no larger than $f(\mathbf{x}^*)$ can equal $f(\mathbf{x}^*)$ only if both equal $f(\mathbf{x}^*)$. Thus $\mathbf{u}$ and $\mathbf{v}$ are also optimal.

If either endpoint is not a vertex, repeat the argument on a larger face. A polygon has finitely many faces, so this process reaches an optimal vertex. The minimum case follows by applying the maximum argument to $-f$.

Detailed graphical maximization example

A fictional Spurs fan-night stand prepares

- x : premium merchandise bundles;
- y : standard merchandise bundles.

The resource constraints are

$$\begin{aligned}
x + y &\leq 8, \\
2x + y &\leq 12, \\
x + 2y &\leq 14, \\
x, y &\geq 0.
\end{aligned}$$

The profit objective is

$$P = 70x + 60y.$$

Using the Section 9.1 vertices:

Vertex	Profit calculation	Profit
$(0, 0)$	$70(0) + 60(0)$	0
$(6, 0)$	$70(6) + 60(0)$	420
$(4, 4)$	$70(4) + 60(4)$	520
$(2, 6)$	$70(2) + 60(6)$	500
$(0, 7)$	$70(0) + 60(7)$	420

Therefore the maximum profit is

$$P_{\max} = 520$$

at

$$(x, y) = (4, 4).$$

Geometric interpretation

Changing objective coefficients changes the slope of the objective level lines while the feasible region remains fixed. The last feasible vertex touched by a moving level line is the optimizer.

Multiple optimal solutions

If an objective level line is parallel to an active boundary edge, the final level line can touch an entire segment. For example, on the feasible region above, maximize

$$Q = x + y.$$

The constraint $x + y \leq 8$ implies $Q \leq 8$. Every feasible point on the edge from $(4, 4)$ to $(2, 6)$ satisfies $x + y = 8$, so every point on that segment is optimal.

The objective value is unique, but the optimizer is not.

Unbounded feasible region versus unbounded objective

An unbounded feasible region does not automatically imply that the objective is unbounded.

- If the improving direction points into an unlimited feasible direction, no finite optimum exists.
- If the objective worsens along every recession direction, a finite optimum may still occur at a vertex.

For example, with $x, y \geq 0$ and $x + y \geq 4$, minimizing $x + y$ has the finite minimum 4, even though the feasible region is unbounded. Maximizing $x + y$ over the same region is unbounded.

Modeling checklist

1. Define each decision variable with units.
2. Write the objective and state maximize or minimize.
3. Translate each resource statement into a constraint.
4. Include nonnegativity restrictions.
5. Draw or compute the feasible region.
6. Find every relevant vertex.
7. Evaluate the objective at each vertex.
8. Interpret the optimizer using the original units.
9. Check every constraint and report slack or surplus.

Section 9.2 terminology

Term	Meaning
Decision variable	A controllable unknown quantity
Objective function	The linear quantity being maximized or minimized
Constraint	A restriction on the decision variables
Feasible solution	A point satisfying every constraint
Optimal solution	A feasible point with the best objective value
Level line	A line on which the objective is constant
Multiple optima	More than one feasible point has the same optimal value
Infeasible problem	No point satisfies every constraint
Unbounded objective	The objective can improve without limit

Section 9.3 - The Simplex Method: Maximization

Learning objectives

Students should be able to:

- convert a maximization problem with $\leq$ constraints to standard form;
- introduce and interpret slack variables;
- identify basic and nonbasic variables;
- construct and read a simplex tableau;
- justify the entering-variable and ratio-test rules;
- perform every Gauss-Jordan pivot exactly;
- recognize optimality, unboundedness, degeneracy, and multiple optima;
- solve a multi-variable maximization problem.

Standard form and slack variables

A maximization problem is in standard form when

$$\max z = \mathbf{c}^T \mathbf{x}$$

subject to

$$A\mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0}, \quad \mathbf{b} \geq \mathbf{0}.$$

For each constraint

$$\mathbf{a}_i^T \mathbf{x} \leq b_i,$$

introduce a slack variable $s_i \geq 0$:

$$\mathbf{a}_i^T \mathbf{x} + s_i = b_i.$$

The slack measures unused capacity.

Basic solutions and vertices

With m independent constraint equations, choose m variables as **basic variables** and set the remaining variables to zero. Solve for the basic variables.

- A **basic solution** is obtained from such a choice.
- A **basic feasible solution** has all variables nonnegative.

Under the usual nondegeneracy assumptions, basic feasible solutions correspond to vertices of the feasible region. The simplex method moves from one basic feasible solution to an adjacent one while improving the objective.

Why the entering variable has the most negative objective-row coefficient

We use the tableau convention

$$z - c_1x_1 - \cdots - c_nx_n = 0.$$

After pivoting to canonical form, the bottom row has the structure

$$z + r_1x_1 + \cdots + r_nx_n = z_0.$$

Equivalently,

$$z = z_0 - r_1x_1 - \cdots - r_nx_n.$$

If a nonbasic variable x_j has $r_j < 0$, then increasing x_j from zero initially changes the objective by

$$-r_jx_j > 0.$$

Thus a negative coefficient indicates a possible improvement. Choosing the most negative coefficient gives the largest immediate improvement per unit of that variable. Other negative choices are mathematically valid, although the number of pivots may differ.

Why the ratio test preserves feasibility

Suppose x_j enters. In canonical form, each basic variable has the form

$$x_{B_i} = b_i - a_{ij}x_j$$

when other nonbasic variables remain zero.

To preserve $x_{B_i} \geq 0$:

- if $a_{ij} > 0$, then

$$x_j \leq \frac{b_i}{a_{ij}};$$

- if $a_{ij} = 0$, the row does not restrict x_j ;
- if $a_{ij} < 0$, increasing x_j increases that basic variable and creates no upper bound.

Therefore the largest feasible increase is

$$\min_{a_{ij} > 0} \frac{b_i}{a_{ij}}.$$

The row attaining this minimum contains the departing variable. At that step, its basic value reaches zero exactly.

Why a nonnegative bottom row proves optimality

At a feasible canonical tableau, suppose

$$z + r_1x_1 + \cdots + r_nx_n = z_0$$

with every reduced coefficient $r_j \geq 0$. Because every nonbasic variable is constrained to be nonnegative,

$$r_1x_1 + \cdots + r_nx_n \geq 0.$$

Hence

$$z = z_0 - \sum_j r_j x_j \leq z_0.$$

The current basic feasible solution achieves z_0 , so no feasible solution can do better. This proves the simplex optimality test.

Detailed simplex example

Maximize

$$z = 3x_1 + 5x_2$$

subject to

$$\begin{aligned} x_1 &\leq 4, \\ x_2 &\leq 6, \\ 3x_1 + 2x_2 &\leq 18, \\ x_1, x_2 &\geq 0. \end{aligned}$$

Add slack variables:

$$\begin{aligned} x_1 + s_1 &= 4, \\ x_2 + s_2 &= 6, \\ 3x_1 + 2x_2 + s_3 &= 18, \\ z - 3x_1 - 5x_2 &= 0. \end{aligned}$$

The initial basic feasible solution is

$$(x_1, x_2, s_1, s_2, s_3) = (0, 0, 4, 6, 18), \quad z = 0.$$

Geometry of a simplex pivot

A basic feasible solution corresponds to a vertex. A pivot changes the basis and typically moves to an adjacent vertex. The entering variable selects an improving direction, and the ratio test identifies the first constraint that becomes tight.

Pivot 1 in full detail

The most negative bottom-row entry is -5 , so x_2 enters. The positive entries in the x_2 column give ratios

$$\frac{6}{1} = 6, \quad \frac{18}{2} = 9.$$

The minimum is 6, so s_2 departs. The pivot is already 1. Apply

$$R_3 \leftarrow R_3 - 2R_2, \quad R_z \leftarrow R_z + 5R_2.$$

The new tableau represents

$$(x_1, x_2, s_1, s_2, s_3) = (0, 6, 4, 0, 6), \quad z = 30.$$

Pivot 2 in full detail

The remaining negative objective-row coefficient is -3 , so x_1 enters. The positive entries in the x_1 column give

$$\frac{4}{1} = 4, \quad \frac{6}{3} = 2.$$

Thus s_3 departs and the pivot is 3. First normalize:

$$R_3 \leftarrow \frac{1}{3}R_3.$$

Then eliminate the other entries in the x_1 column:

$$R_1 \leftarrow R_1 - R_3, \quad R_z \leftarrow R_z + 3R_3.$$

The final bottom row is nonnegative. Setting nonbasic variables $s_2 = s_3 = 0$ gives

$$x_1 = 2, \quad x_2 = 6, \quad s_1 = 2.$$

Therefore

$$z_{\max} = 3(2) + 5(6) = 36.$$

Interpreting the final solution

At $(x_1, x_2) = (2, 6)$:

$$x_1 = 2 < 4,$$

so the first constraint has slack $s_1 = 2$.

The other two constraints are active:

$$x_2 = 6,$$

and

$$3x_1 + 2x_2 = 3(2) + 2(6) = 18.$$

A positive slack means unused capacity. A zero slack means the corresponding resource is fully used.

Three-decision-variable extension

Maximize

$$z = 4x_1 + 3x_2 + 2x_3$$

subject to

$$\begin{aligned}x_1 + x_2 + x_3 &\leq 4, \\2x_1 + x_2 &\leq 5, \\x_2 + 2x_3 &\leq 6, \\x_1, x_2, x_3 &\geq 0.\end{aligned}$$

The simplex method is unchanged. The tableau simply has one additional decision-variable column. The final basic solution is

$$x_1 = \frac{5}{2}, \quad x_2 = 0, \quad x_3 = \frac{3}{2},$$

with

$$z = 4 \left(\frac{5}{2} \right) + 2 \left(\frac{3}{2} \right) = 13.$$

The fractions are exact; no rounding is necessary.

Exceptional simplex outcomes

Unbounded objective

If an entering column has no positive constraint entry, the entering variable can increase indefinitely without forcing a basic variable below zero. The maximization problem is unbounded.

Degeneracy

If the minimum ratio is zero, the next basic feasible solution has a basic variable equal to zero. The objective may fail to improve during that pivot. Degeneracy can cause repeated bases; Bland's rule prevents cycling by choosing the smallest-index eligible variables.

Multiple optima

If a nonbasic decision variable has zero reduced cost in a final tableau, it can enter without changing the objective. This indicates an alternate optimal basic feasible solution and usually an entire optimal edge.

Infeasibility

If no initial basic feasible solution exists, Phase I with artificial variables is required. A positive Phase I optimum proves that the original constraints are infeasible.

Section 9.3 terminology

Term	Meaning
Standard form	Maximization with $A\mathbf{x} \leq \mathbf{b}$, $\mathbf{x} \geq 0$, $\mathbf{b} \geq 0$
Slack variable	Unused amount in a $\leq$ constraint
Basic variable	A variable represented by a pivot column
Nonbasic variable	A variable set to zero at a basic solution
Basic feasible solution	A nonnegative basic solution

Term	Meaning
Entering variable	A nonbasic variable selected to improve the objective
Departing variable	The basic variable forced to zero by the ratio test
Pivot	The entry used for a Gauss-Jordan basis change
Reduced cost	Objective-row coefficient after canonicalization
Degeneracy	A basic feasible solution with a zero basic variable

Section 9.4 - The Simplex Method: Minimization and Duality

Learning objectives

Students should be able to:

- write a minimization problem in standard form;
- construct its dual maximization problem;
- prove and apply weak duality;
- understand the simplex-basis proof of strong duality;
- solve a minimization problem by solving its dual;
- use complementary slackness to connect primal and dual optimizers;
- interpret dual variables as marginal resource values.

Standard primal minimization and dual maximization

Consider the minimization problem

$$\begin{array}{ll}\min & w = \mathbf{c}^T \mathbf{x} \\ \text{subject to} & A\mathbf{x} \geq \mathbf{b}, \\ & \mathbf{x} \geq \mathbf{0}.\end{array}$$

Its dual is

$$\begin{array}{ll}\max & z = \mathbf{b}^T \mathbf{y} \\ \text{subject to} & A^T \mathbf{y} \leq \mathbf{c}, \\ & \mathbf{y} \geq \mathbf{0}.\end{array}$$

Rows of the primal become variables of the dual, columns of the primal become constraints of the dual, and the coefficient matrix is transposed.

Weak duality theorem

Weak duality

For every primal-feasible $\mathbf{x}$ and dual-feasible $\mathbf{y}$,

$$\mathbf{b}^T \mathbf{y} \leq \mathbf{c}^T \mathbf{x}.$$

Thus every dual value is a lower bound for every primal minimization value.

Complete proof of weak duality

Primal feasibility gives

$$A\mathbf{x} \geq \mathbf{b}.$$

Because $\mathbf{y} \geq \mathbf{0}$, multiplying componentwise and summing preserves the inequality:

$$\mathbf{y}^T A\mathbf{x} \geq \mathbf{y}^T \mathbf{b} = \mathbf{b}^T \mathbf{y}.$$

Dual feasibility gives

$$A^T \mathbf{y} \leq \mathbf{c}.$$

Because $\mathbf{x} \geq \mathbf{0}$,

$$\mathbf{x}^T A^T \mathbf{y} \leq \mathbf{x}^T \mathbf{c} = \mathbf{c}^T \mathbf{x}.$$

The middle scalar is the same in both expressions:

$$\mathbf{y}^T A \mathbf{x} = \mathbf{x}^T A^T \mathbf{y}.$$

Therefore

$$\boxed{\mathbf{b}^T \mathbf{y} \leq \mathbf{c}^T \mathbf{x}}.$$

Strong duality through an optimal simplex basis

For a primal maximization problem

$$\max \mathbf{c}^T \mathbf{x} \quad \text{subject to} \quad A \mathbf{x} \leq \mathbf{b}, \quad \mathbf{x} \geq 0,$$

suppose an optimal basis matrix B has been found. The basic solution is

$$\mathbf{x}_B = B^{-1} \mathbf{b}.$$

Define

$$\mathbf{y}^T = \mathbf{c}_B^T B^{-1}.$$

At an optimal tableau, every reduced cost satisfies

$$\mathbf{c}_N^T - \mathbf{y}^T N \leq 0,$$

so

$$A^T \mathbf{y} \geq \mathbf{c}.$$

The slack-variable columns are identity columns. Their reduced costs are $-y_i \leq 0$, so $\mathbf{y} \geq 0$. Thus $\mathbf{y}$ is dual feasible. Finally,

$$\mathbf{b}^T \mathbf{y} = \mathbf{c}_B^T B^{-1} \mathbf{b} = \mathbf{c}_B^T \mathbf{x}_B.$$

The primal and dual feasible values are equal. Weak duality says neither can be improved, so both are optimal. This is the simplex-basis proof of strong duality.

Detailed minimization example

Minimize

$$w = 6x_1 + 4x_2$$

subject to

$$\begin{aligned}x_1 + x_2 &\geq 5, \\2x_1 + x_2 &\geq 8, \\x_1, x_2 &\geq 0.\end{aligned}$$

The dual is

$$\max z = 5y_1 + 8y_2$$

subject to

$$\begin{aligned}y_1 + 2y_2 &\leq 6, \\y_1 + y_2 &\leq 4, \\y_1, y_2 &\geq 0.\end{aligned}$$

Dual pivot calculations

Initially y_2 enters because -8 is most negative. Ratios are

$$\frac{6}{2} = 3, \quad \frac{4}{1} = 4.$$

Thus s_1 departs. After dividing the first row by 2 and eliminating the y_2 column, y_1 still has a negative reduced cost. The next ratios are

$$\frac{3}{1/2} = 6, \quad \frac{1}{1/2} = 2.$$

Thus s_2 departs. The final solution is

$$y_1 = 2, \quad y_2 = 2, \quad z = 26.$$

By strong duality,

$$\boxed{w_{\min} = 26}.$$

The coefficients under the dual slack columns in the final objective row recover the primal solution:

$$\boxed{x_1 = 3, \quad x_2 = 2}.$$

Check:

$$6(3) + 4(2) = 26.$$

Complementary slackness

For the primal minimization and dual maximization pair,

$$x_j (c_j - (A^T \mathbf{y})_j) = 0$$

for every primal variable, and

$$y_i ((A\mathbf{x})_i - b_i) = 0$$

for every primal constraint.

Why it follows

Weak duality can be written as

$$\mathbf{c}^T \mathbf{x} - \mathbf{b}^T \mathbf{y} = \mathbf{x}^T (\mathbf{c} - A^T \mathbf{y}) + \mathbf{y}^T (A\mathbf{x} - \mathbf{b}).$$

Every component in both dot products is nonnegative for feasible $\mathbf{x}$ and $\mathbf{y}$. At optimality, strong duality makes the left side zero. A sum of nonnegative terms can be zero only when every term is zero. This yields the complementary-slackness equations.

Complementary-slackness check for the example

The primal solution is $\mathbf{x} = (3, 2)$ and the dual solution is $\mathbf{y} = (2, 2)$.

Both primal constraints are active:

$$3 + 2 = 5,$$

$$2(3) + 2 = 8.$$

Both dual constraints are also active:

$$2 + 2(2) = 6,$$

$$2 + 2 = 4.$$

Because all four variables are positive, complementary slackness correctly requires all four associated constraints to be active.

Economic interpretation of dual variables

A dual variable is often called a **shadow price**. It estimates how the optimal objective changes when the corresponding right-hand side changes by one unit while the current optimal basis remains valid.

In the example, $y_1 = 2$ and $y_2 = 2$. Increasing either minimum requirement by one unit raises the minimum cost by approximately two objective units within the allowable sensitivity range.

This interpretation is local: a sufficiently large change can alter the optimal basis and hence the shadow prices.

Section 9.4 terminology

Term	Meaning
Primal problem	The original optimization problem
Dual problem	The transposed companion optimization problem
Weak duality	Every dual feasible value bounds every primal feasible value
Strong duality	Finite primal and dual optima have equal values
Dual variable	A variable associated with one primal constraint
Shadow price	Marginal value of relaxing or tightening a constraint
Complementary slackness	Product conditions linking positive variables and active constraints
Duality gap	Primal value minus dual value; zero at optimality

Section 9.5 - Mixed Constraints and the Two-Phase Simplex Method

Learning objectives

Students should be able to:

- distinguish slack, surplus, and artificial variables;
- explain why a $\geq$ constraint does not provide an obvious basic variable;
- formulate and solve the Phase I auxiliary problem;
- prove the Phase I feasibility criterion;
- remove artificial variables and restore the original objective in Phase II;
- convert mixed minimization to maximization when appropriate;
- recognize infeasibility from a positive Phase I optimum;
- formulate transportation and allocation applications.

Slack, surplus, and artificial variables

For

$$\mathbf{a}^T \mathbf{x} \leq b,$$

add a slack variable:

$$\mathbf{a}^T \mathbf{x} + s = b, \quad s \geq 0.$$

For

$$\mathbf{a}^T \mathbf{x} \geq b,$$

subtract a surplus variable:

$$\mathbf{a}^T \mathbf{x} - s = b, \quad s \geq 0.$$

The coefficient of s is -1 , so it does not form an identity column for an initial basis. Add an artificial variable:

$$\mathbf{a}^T \mathbf{x} - s + a = b, \quad a \geq 0.$$

The artificial variable is temporary and must equal zero before solving the original problem.

Phase I and Phase II

Phase I

Minimize the sum of all artificial variables:

$$w = a_1 + \cdots + a_k.$$

Equivalently, maximize $-w$. Pivot until the auxiliary objective is optimal.

Phase II

If the Phase I minimum is zero:

1. remove all artificial-variable columns;
2. remove any redundant zero rows;
3. restore the original objective;
4. eliminate coefficients of current basic variables from the objective row;
5. continue ordinary simplex pivots.

Proof of the Phase I feasibility criterion

Every artificial variable satisfies $a_i \geq 0$, so

$$w = a_1 + \cdots + a_k \geq 0.$$

- If Phase I attains $w = 0$, then every $a_i = 0$. The remaining original, slack, and surplus variables satisfy the original constraints, so the original problem is feasible.
- Conversely, if the original problem has a feasible solution, set every artificial variable to zero. This gives a feasible Phase I solution with $w = 0$. Therefore the Phase I minimum must be zero.
- If the Phase I minimum is positive, no feasible point can make all artificial variables zero. Hence the original constraints are infeasible.

Thus

$\text{original problem feasible} \iff w_{\min} = 0.$

Detailed two-phase example

Maximize

$$z = 5x_1 + 4x_2$$

subject to

$$\begin{aligned} x_1 + x_2 &\leq 6, \\ 2x_1 + x_2 &\geq 4, \\ x_1, x_2 &\geq 0. \end{aligned}$$

Introduce variables:

$$\begin{aligned} x_1 + x_2 + s_1 &= 6, \\ 2x_1 + x_2 - s_2 + a_2 &= 4. \end{aligned}$$

For Phase I, maximize $-w = -a_2$. Because a_2 is initially basic, eliminate it from the objective row by subtracting the second constraint row.

Phase I pivot in detail

The Phase I bottom row has -2 under x_1 and -1 under x_2 , so choose x_1 to enter. Ratios are

$$\frac{6}{1} = 6, \quad \frac{4}{2} = 2.$$

Thus the artificial variable a_2 departs. Normalize and eliminate:

$$R_2 \leftarrow \frac{1}{2}R_2,$$

$$R_1 \leftarrow R_1 - R_2, \quad R_w \leftarrow R_w + 2R_2.$$

The Phase I objective value is zero and a_2 is nonbasic. The original problem is feasible.

Phase II pivots in detail

After restoring the original objective and eliminating the basic x_1 coefficient, the tableau is

	x_1	x_2	s_1	s_2	b
R_1	0	$\frac{1}{2}$	1	$\frac{1}{2}$	4
R_2	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	2
R_z	0	$-\frac{3}{2}$	0	$-\frac{5}{2}$	10.

Any negative reduced cost can be selected. To illustrate this flexibility, we choose x_2 even though s_2 has the most negative coefficient.

Pivot 1: x_2 enters

The eligible ratios are

$$\frac{4}{1/2} = 8, \quad \frac{2}{1/2} = 4.$$

Thus x_1 leaves. Apply

$$R_2 \leftarrow 2R_2, \quad R_1 \leftarrow R_1 - \frac{1}{2}R_2, \quad R_z \leftarrow R_z + \frac{3}{2}R_2.$$

The result is

	x_1	x_2	s_1	s_2	b
R_1	-1	0	1	1	2
R_2	2	1	0	-1	4
R_z	3	0	0	-4	16.

Pivot 2: s_2 enters

Only R_1 has a positive entry in the s_2 column, so s_1 leaves. Apply

$$R_2 \leftarrow R_2 + R_1, \quad R_z \leftarrow R_z + 4R_1.$$

Then

	x_1	x_2	s_1	s_2	b
R_1	-1	0	1	1	2
R_2	1	1	1	0	6
R_z	-1	0	4	0	24.

Pivot 3: x_1 enters

Only R_2 has a positive entry in the x_1 column, so x_2 leaves. Apply

$$R_1 \leftarrow R_1 + R_2, \quad R_z \leftarrow R_z + R_2.$$

The final tableau is

	x_1	x_2	s_1	s_2	b
R_1	0	1	2	1	8
R_2	1	1	1	0	6
R_z	0	1	5	0	30.

The nonbasic variables are $x_2 = s_1 = 0$, so

$$x_1 = 6, \quad s_2 = 8, \quad \boxed{z_{\max} = 30}.$$

The final row gives

$$z + x_2 + 5s_1 = 30,$$

which proves $z \leq 30$ for every feasible solution.

Mixed minimization by objective negation

To minimize

$$w = \mathbf{c}^T \mathbf{x},$$

one may maximize

$$z = -\mathbf{c}^T \mathbf{x}.$$

The feasible set is unchanged, and

$$w_{\min} = -z_{\max}.$$

Mixed $\geq$, $\leq$, and equality constraints still require the same surplus/artificial-variable treatment. Objective negation changes only the objective row, not the constraint equations.

Transportation-style application

A fictional travel-equipment distributor ships equipment from two depots to two hiking centers. After using demand equations to eliminate two shipment variables, the model can produce both capacity constraints of type $\leq$ and demand constraints of type $\geq$.

A complete model should include:

- supply limits at each depot;
- minimum demand at each center;
- nonnegative shipment quantities;
- a linear shipping-cost objective.

Transportation problems have additional structure and can be solved by specialized algorithms, but the two-phase simplex method remains valid.

A concrete two-depot/two-center model uses shipment variables $x_{ij} \geq 0$ with supply constraints

$$x_{11} + x_{12} \leq S_1, \quad x_{21} + x_{22} \leq S_2,$$

demand requirements

$$x_{11} + x_{21} \geq D_1, \quad x_{12} + x_{22} \geq D_2,$$

and linear shipping cost

$$\min c_{11}x_{11} + c_{12}x_{12} + c_{21}x_{21} + c_{22}x_{22}.$$

Section 9.5 terminology

Term	Meaning
Surplus variable	Amount by which a $\geq$ requirement is exceeded
Artificial variable	Temporary variable used to create an initial basis
Phase I	Auxiliary optimization that searches for feasibility
Phase II	Optimization of the original objective after feasibility is found
Big-M method	Alternative method assigning a large penalty to artificial variables
Mixed constraints	A model containing different inequality directions or equalities
Infeasible tableau	Phase I ends with positive artificial-variable sum
Redundant constraint	A constraint that does not change the feasible set

The **Big-M method** is an alternative to two-phase simplex. Artificial variables are kept in the original objective but assigned a very large penalty M . This course emphasizes two-phase simplex because it separates feasibility from optimization more cleanly.

Chapter 9 synthesis

Linear programming links geometry, algebra, and algorithms:

1. Linear inequalities create convex feasible regions.
2. Linear objectives attain finite optima at vertices.
3. Basic feasible solutions are the algebraic representation of those vertices.
4. A simplex pivot moves from one basis to an adjacent basis.
5. Reduced costs determine whether improvement is possible.
6. Duality gives certificates and bounds for optimality.
7. Complementary slackness identifies which variables and constraints can be simultaneously positive or slack.
8. Artificial variables turn feasibility itself into an optimization problem.

The simplex method is therefore not a collection of arbitrary tableau rules. Every step follows from feasibility, linearity, convexity, and exact row-equivalent reformulation.

Connection to earlier linear algebra

Linear programming is not separate from the rest of the course. The feasible set is built from linear equations and inequalities, simplex tableaux use row operations, basic variables correspond to selected columns, and duality compares two related linear systems. The new ingredient is optimization: among all feasible solutions, we choose one that maximizes or minimizes a linear objective.